

Alannah “Al” Oleson, Ph.D.

Assistant Professor, Computer Science
University of Denver

OlesonPhD@gmail.com
alannah.oleson@du.edu
alannaholeson.com
[linkedIn] @AOleson

Research Overview

I envision a world where *all* computing students can create technology that is accessible, inclusive, and ethical, because their education equips them with the ability to critically reason about the real-world impacts of their technology design decisions on end users and society.

My research explores teacher knowledge, CS disciplinary norms, and pedagogical methods for justice-centered software design & development, including AI design and use, in real-world post-secondary learning contexts. I use qualitative and participatory action research methods, applying statistical quantitative analyses when needed. My work sits in the intersection of human-computer interaction and computing education research and is heavily informed by design justice, disability justice, and liberatory pedagogies.

Employment

- Sept. 2024 **Assistant Professor**
-present Ritchie School of Engineering and Computer Science, University of Denver
Director of TEACUP (Teaching Empathy, Agency, and Criticality for Unlearning Power) Lab
- 2023-2024 **Postdoctoral Scholar & Co-Founder**
Center for Learning, Computing, and Imagination, University of Washington
- 2018-2023 **Graduate Research Assistant**
Code & Cognition Lab, University of Washington (Advisor: Amy J. Ko)
- 2017-2018 **Procedural Imaging Group Intern**
Adobe Research (Mentors: Cynthia Lu, José Echevarria, Radomir Mech)
- 2014-2018 **Undergraduate Research Assistant**
EUSES / Gender HCI Lab, Oregon State University (Advisor: Margaret M. Burnett)

Research Interests

HCI Education, Computing Education, Human-Computer Interaction, Design Justice, User Experience Design, Critical Computing, AI Education, Human-Centered AI, Responsible AI, Inclusive Design, Ethical Computing

Education

- 2018-2023 **University of Washington**, Seattle, WA, USA
Ph.D. in Information Science (Advisor: Amy J. Ko)
Dissertation: *Integrating Inclusive Design and Computing Education*
- 2021 **University of Washington**, Seattle, WA, USA
M.S. in Information Science (Advisor: Amy J. Ko)
Thesis: *Toward Justice-Centered Software Design & Development*
- 2014-2018 **Oregon State University**, Corvallis, OR, USA
Honors B.S. in Computer Science, summa cum laude (Advisor: Margaret M. Burnett)
Thesis: *Pedagogical Content Knowledge for Teaching Inclusive Software Design*

Awards & Honors

- 2026 **Faculty Career Champion Award of Excellence**
University of Denver
Awarded by student nomination & career council review to a faculty member who has gone above and beyond to support students' career, internship, and post-graduation goals.
- Provost's Award for Excellence in Academic Advising**
University of Denver
Awarded by student nomination for creating conditions where all students can thrive.
- 2025 **Dean's Award**, Ritchie School of Engineering and Computer Science
University of Denver
Awarded in first year of assistant professorship for significant contributions to the school's research goals, teaching & mentorship programs, and service to the DU community.
- Excellence in Undergraduate Mentoring Award**, Student Signature Work Program
University of Denver
- 2023 **Best Paper Award**, ACM ICER
- 2022 **Best Paper Award for Diversity & Inclusion Contribution**, ACM CSCW
- 2022 **Husky 100**
University of Washington
Awarded annually to 100 of UW's 60,000 students who make exceptional contributions to research, teaching, mentorship, and service to the institution. Highest honor available for students.
- 2018 **Outstanding Undergraduate Researcher, Finalist**, Computing Research Association (CRA)
- 2017 **Best Paper Honorable Mention**, ACM CHI
- 2015, 2016 **Drucilla Shepard Smith Academic Excellence Award**, Oregon State University

External Research Funding

- 2026-2029 **National Science Foundation**
Collaborative Research: Multisite Comparison of Embedding Socially Responsible Design Across the Undergraduate Computer Science Curriculum
PI: **Alannah Oleson**; Collaborative proposal with 4 other universities (Lead: Patricia Morreale (Kean); Institutional PIs: Margaret Burnett, Christopher Hundhausen (Oregon State), Briana Morrison (UVA))
NSF DUE IUSE 23-510: Improving Undergraduate STEM Education Level 2 (Institutional Transformation)
\$400,942 (\$165,951 for U. Denver) — Award # 2539896
- 2026-2028 **Mozilla Foundation: Mozilla Responsible Computing Challenge**
CHAI: Career-Ready Humanities-based AI Education Modules
PI: **Alannah Oleson**; Co-PI: Benjamin Xie (U. Denver)
DU Faculty Partners: Lindsey Feitz (Gender & Women's Studies), Kaitlyn Sims (Public Policy), Laurel Taylor (Japanese Studies)
\$125,000
- 2025-2030 **National Science Foundation**
RELATE Scholars: Relationships Elevating Students to Achieve Transformational Educations
PI: Rachel Horenstein; Co-Is: **Alannah Oleson**; Benjamin Xie, Michelle Sabick, Alex Stevens (U. Denver)
NSF 25-514: NSF Scholarships in Science, Technology, Engineering, and Mathematics Program (S-STEM)
\$996,598 — Award # 2424972

- 2026 **Uppsala University - Equal Opportunities Group**
Visiting Researcher Grant
 Equal Opportunities Group, Faculty of Science & Technology
 50,000kr (~\$5,250)
- 2025-2026 **Research Innovation Scotland (RIS) International Collaboration Fund**
Computing Students' Perceptions of their Own Ethical Agency in Technology Design Processes
 PIs: **Alannah Oleson** (U. Denver), Matthew Barr, Maria Kallia (U. Glasgow), Bedour Alshaigy (Uppsala U.)
 £4,520 (~\$5,940)
- 2025 **Uppsala University - Equal Opportunities Group**
Visiting Researcher Grant
 Equal Opportunities Group, Faculty of Science & Technology
 50,000kr (~\$5,250)
- 2018-2023 **National Science Foundation**
Graduate Research Fellowship (GRFP)
 Comp/IS/Eng - Human Computer Interaction
 \$138,000
- 2017 **Adobe Research**
Women in Tech Undergraduate Scholar
 \$10,000

Internal Research Funding

- 2026-2027 **Center for Ethical Generative Artificial Intelligence Applications**, U. Denver
Measuring Semantic Fidelity in LLM-Based Privacy Policy Interpretation for Older Adults
 PI: **Alannah Oleson**; Co-Is: Iffat Anjum, Suleiman Saka (U. Denver)
 \$7,000
- 2026 **Discovery PINS Student Signature Work Grant**, U. Denver
Exploring Accessible Teaching Practices in Computer Science Syllabi
 Mentor: **Alannah Oleson**; Undergraduate Researcher: Muskan Fatima
 \$3,800
- 2025 **Discovery PINS Student Signature Work Grant**, U. Denver
Evaluating a Pedagogical Resource for Teaching Accessible Computing
 Mentor: **Alannah Oleson**; Undergraduate Researcher: Muskan Fatima
 \$1,500

Invited Talks

- 2026 ***Teaching Accessible Computing: A Resource for CS Faculty***
 @ AccessComputing Workshop, ACM SIGCSE Technical Symposium
- How Do We Educate Compassionate Citizens?***
 @ U. Denver 4D Experience Kickoff: Compassion Lab Golden Hour
- 2025 ***Toward Critical, Ethical, Creative, and Reflective HCI Education Futures***
 @ ATLAS Colloquium, University of Colorado - Boulder
- Re-Imagining Academia as Collaborative and Community-Centered***
 @ Community Dinner Series, Equity In STEM (E-STEM) Program at U. Denver

- 2024 **Doing HCI Education Research in Your Own Classroom**
@ Pedagogy Workshop, EduCHI Annual Symposium on HCI Education
- 2021 **Panel: Can Researchers Enhance Diversity and Inclusion?**
@ International Conference on Software Engineering (ICSE)
Recording available at <https://youtu.be/6LROK9Zqiks>
- 2018 **Research Symposium Welcome Address**
@ Celebrating Undergraduate Excellence (CUE) Event, Oregon State University

Note on Publication Strategy and Authorship

In Computer Science and related fields, peer-reviewed conference proceedings serve as the primary venue for disseminating novel research contributions, functioning in a role comparable to journals in other disciplines. Many leading conferences employ rigorous review processes, low acceptance rates, and archival proceedings. As a result, conference publications are major scholarly outputs within the field.

My publication strategy reflects disciplinary norms in Computer Science while also engaging with interdisciplinary audiences through journal, book, and workshop publications. Further, I engage in small and emergent communities that represent the forefront of novel scholarship, which may have higher acceptance rates. I also practice public scholarship by editing and contributing to multiple “living document” digital textbooks for computing educators, making scholarly knowledge more accessible.

As academics become more established in my field, the publishing convention is to begin to prioritize anchor/last authorship positions, which indicate a leadership role on the paper. This is particularly true for work led by my laboratory/advisees and students in other labs that I mentor through collaborative research projects.

Publications

My publications have been cited ~1,300 times, and I have an h-index of 16 ([Google Scholar](#), Sept. 2026). Underlined authors below indicate students and junior scholars that I advised and/or mentored.

Journal Articles

5-year impact factor (IF) provided when available. 5-year IF preferred because yearly IFs fluctuate considerably.

- [j7] (In draft) **Integrating Critical HCI Topics into Computing Education**
A. Oleson et al. (2026)
ACM TOCHI: Transactions on Computer-Human Interaction (IF = 9.8)
- [j6] **Deconstructing Conceptions of Rigor in Computer Science Education**
J. Everson, F.M. Kivuva, E. Sherif, **A. Oleson**, A.J. Ko (2025)
ACM TOCE: Transactions on Computing Education (IF = 4.6)
- [j5] **Factors Influencing the Social Help-Seeking Behavior of Introductory Computing Students in a Competitive University Environment**
A.K. Cohen, **A. Oleson**, A.J. Ko (2024)
ACM TOCE: Transactions on Computing Education (IF = 4.6)
- [j4] **Teaching Inclusive Design Skills with the CIDER Assumption Elicitation Technique**
A. Oleson, M. Solomon, C. Perdriau, A.J. Ko (2023)
ACM TOCHI: Transactions on Computer-Human Interaction (IF = 9.8)

- [j3] ★ Best Paper Award for Diversity & Inclusion Contribution
Surfacing Equity Issues in Large Computing Courses with Peer-Ranked, Demographically-Labeled Student Feedback
 B. Xie, **A. Oleson**, J. Everson, A.J. Ko (2022)
 PACMHCI: Proceedings of the ACM on Human-Computer Interaction (CSCW) (IF = 4.9)
- [j2] **On the Role of Design in K-12 Computing Education**
A. Oleson, B. Wortzman, A.J. Ko (2021)
 ACM TOCE: Transactions on Computing Education (IF = 4.6)
- [j1] **General Principles for a Generalized Idea Garden**
 W. Jernigan, A. Horvath, M. Lee, M.M. Burnett, T. Culty, S. Kuttal, A. Peters, I. Kwan, F. Bahmani, A.J. Ko, C.J. Mendez, **A. Oleson** (2017)
 Journal of Visual Languages and Computing (IF = 1.2)

Conference Publications (Strictly Peer-Reviewed)

Acceptance rates (AR) provided where available.

- [c18] (Under review) **"I See Myself at Every Step": Computing Students' Conceptions of Their Own Roles And Responsibilities in Human-Centered Design**
A. Oleson, B. Alshaigy (2026)
 ACM CHI: Conference on Human Factors in Computing Systems
- [c17] (Under review) **Beyond One-Size-Fits-All: Privacy, Security, and Stigma in Telehealth Through the Lens of Marginalized Users**
F. Tazi, S. Saka, Y. Kim, **A. Oleson** (2026)
 ACM CHI: Conference on Human Factors in Computing Systems
- [c16] (Under review) **Measuring Semantic Fidelity in LLM-Based Privacy Policy Interpretation for Older Adults**
S. Saka, **A. Oleson**, I. Anjum (2026)
 ACM CHI: Conference on Human Factors in Computing Systems
- [c15] (Under review) **Teaching Accessible Computing: Design & Evaluation of a Community-Sourced Digital Textbook**
A. Oleson, M. Fatima, Y. Kim (2026)
 ACM CHI: Conference on Human Factors in Computing Systems
- [c14] (In draft) **From Diary to Discussion: Scaffolding Empathy-Oriented Learning in HCI Education**
J. Fan, **A. Oleson**, S. Edwards, M. Pérez-Quñones, W. Wang, S. McCrickard (2026)
 ACM ITiCSE: Innovation and Technology in Computer Science Education
- [c13] **First Impressions Matter: How CS Syllabi Communicate and Reflect Accessibility Norms**
M. Fatima, **A. Oleson** (2026)
 ACM SIGCSE Technical Symposium
- [c12] **Visualizing Human-Centered Design: A Replication and Extension Study**
A. Oleson, Y. Kim (2026)
 EduCHI Annual Symposium on HCI Education (AR = 37%)
<https://doi.org/10.1145/3803869.3803881>

- [c11] **Starting with DEI and Ethics: A New First-Year Computer Science Introduction**
S. Leutenegger, S. Hutt, A. Hannum, S. Das, **A. Oleson**, A. Leto, S. Shrestha (2026)
ACM SIGCSE Technical Symposium (AR = 30%)
<https://doi.org/10.1145/3770762.3772628>
- [c10] **Design as Translation, Translation as Design: Toward Critical, Creative, and Ethical Pedagogies**
A. Oleson, L. Taylor (2025)
EduCHI Annual Symposium on HCI Education
<https://doi.org/10.1145/3742901.3742902>
- [c9] ★ Best Paper Award
Funds of Knowledge Used by Adolescents of Color in Scaffolded Sensemaking around Algorithmic Fairness
J. Salac, **A. Oleson**, L. Armstrong, A. Le Meur, A.J. Ko (2023)
ACM ICER: International Computing Education Research Conference (AR = 21%)
<https://doi.org/10.1145/3568813.3600110>
- [c8] **A Decade of Demographics in Computing Education Research: A Critical Review of Trends in Collection, Reporting, and Use**
A. Oleson*, B. Xie*, J. Salac, J. Everson, F.M. Kivuva, A.J. Ko (2022)
ACM ICER: International Computing Education Research Conference (AR = 16%)
<https://doi.org/10.1145/3501385.3543967>
- [c7] **Computing Students' Learning Difficulties in HCI Education**
A. Oleson, M. Solomon, A.J. Ko (2020)
ACM CHI: Conference on Human Factors in Computing Systems (AR = 24%)
<https://doi.org/10.1145/3313831.3376149>
- [c6] **Scout: Rapid Exploration of Interface Layout Variations through High-Level Design Constraints**
A. Swearngin, C. Wang, **A. Oleson**, A.J. Ko, J. Fogarty (2020)
ACM CHI: Conference on Human Factors in Computing Systems (AR = 24%)
<https://doi.org/10.1145/3313831.3376593>
- [c5] **Pedagogical Content Knowledge for Teaching Inclusive Design**
A. Oleson, C.J. Mendez, Z. Steine-Hanson, C. Hilderbrand, C. Perdriau, M.M. Burnett, A.J. Ko (2018)
ACM ICER: International Computing Education Research Conference (AR = 22%)
<https://doi.org/10.1145/3230977.3230998>
- [c4] **Semi-Automating (or not) a Socio-Technical Method for Socio-Technical Systems**
C.J. Mendez, Z. Steine-Hanson, **A. Oleson**, A. Horvath, C.G. Hill, C. Hilderbrand, A. Sarma, M.M. Burnett (2018)
IEEE VL/HCC: Symposium on Visual Languages and Human-Centric Computing (AR = 29%)
[10.1109/VLHCC.2018.8506514](https://doi.org/10.1109/VLHCC.2018.8506514)

- [c3] **★ Best Paper Honorable Mention**
Gender-Inclusiveness Personas vs. Stereotyping: Can We Have it Both Ways?
 C.G. Hill, M. Haag, **A. Oleson**, C.J. Mendez, N. Marsden, A. Sarma, M.M. Burnett (2017)
 ACM CHI: Conference on Human Factors in Computing Systems (AR = 25%)
<https://doi.org/10.1145/3025453.3025609>
- [c2] **GenderMag Experiences in the Field: The Whole, the Parts, and the Workload**
 C.G. Hill, S. Ernst, **A. Oleson**, A. Horvath, M.M. Burnett (2016)
 IEEE VL/HCC: Symposium on Visual Languages and Human-Centric Computing
[10.1109/VLHCC.2016.7739685](https://doi.org/10.1109/VLHCC.2016.7739685)
- [c1] **Programming, Problem Solving, and Self-Awareness: Effects of Explicit Guidance**
 D. Loksa, A.J. Ko, W. Jernigan, **A. Oleson**, C.J. Mendez, M.M. Burnett (2016)
 ACM CHI: Conference on Human Factors in Computing Systems (AR = 23%)
<https://doi.org/10.1145/2858036.2858252>

Book Chapters

- [b2] **CIDER Assumption Elicitation Technique**
 Spread in *Universal Methods for Ethical Design: 100 Ways to Become More Ethically Aware, Responsible, and Active in Your Design Work* (2024)
 Editors: S.S. Chivukula, C. Gray (Quarto Publishing)
<https://www.quarto.com/books/9780760393086/universal-methods-of-ethical-design>
- [b1] **Toward Theory-Based End-User Software Engineering**
 M.M. Burnett, T. Kulesza, **A. Oleson**, S. Ernst, L. Beckwith, J. Cao, W. Jernigan, W. Grigoreanu
 Chapter in *New Perspectives in End-User Development* (2017)
 Editors F. Paternò, V. Wulf (Springer Int'l Publishing)
<https://doi.org/10.1007/978-3-319-60291-2>

Professional Magazine Articles (Editor-reviewed)

5-year impact factor (IF) provided when available. 5-year IF preferred because yearly IFs fluctuate considerably.

- [a2] (under review) **AI Policy as a Pragmatic Pathway for Computing Ethics Education**
S. Shrestha, C. Ke, **A. Oleson**, B. Xie (2026)
 ACM Inroads: Advancing Computing Education on a Global Scale (IF = 1.3)
- [a1] **It Is Time for More Critical CS Education**
 A.J. Ko, **A. Oleson**, M. Kirdani-Ryan, Y. Register, B. Xie, M. Tari, M.J. Davidson, S. Druga, D. Loksa (2020)
 ACM CACM: Communications of the ACM (IF = 7.1)
<https://doi.org/10.1145/3424000>

Co-organized Workshops and Panels

- [w8] **Teaching & Learning for Regeneration & Resilience**
 S. Bexell (co-moderator), J. Senecal (co-moderator), **A. Oleson**, S. Chatfield, J. Brown
 Center for Regenerative Futures @ U. Denver

- [w7] **Teach-in & Community Dinner: Sociotechnical Risks & Harms of AI Technologies**
J. Senecal (moderator), A. Espino, **A. Oleson**, S. Sarantakos, D. Sundaram
Graduate School of Social Work & Center for Regenerative Futures @ U. Denver
- [w6] **Accessibility Research and Making Research Accessible**
B. Bettin (moderator), **A. Oleson**, Y. Rankin, R. Kushalnagar, (2025)
ACM RESPECT Doctoral Consortium
- [w5] **Navigating Graduate School**
B. Bettin (moderator), **A. Oleson**, Y. Rankin, R. Kushalnagar, (2025)
ACM RESPECT Doctoral Consortium
- [w4] **Succeeding in the Academic Job Market**
P. Kinnunen (moderator), M. Paretti, B. Alshaigy, A. Eckerdal, **A. Oleson** (2025)
Humanization of Computing and Engineering Education (HCEE) Doctoral Consortium
- [w3] **AiiCE Lecture: Teaching Accessible Computing: A Panel Discussion**
A. Oleson (moderator), C. Baker, W. Barres, E. Kalodner-Martin, T. Pickering, B. Blaser (2025)
Identity & Computing Lecture Series @ Alliance for Interdisciplinary Innovation in Computing Education (AiiCE)
<https://aiiceinnovates.org/event/aiice-lecture-series-teaching-accessible-computing-panel-discussion/>
- [w2] **Including Accessibility in Your Computer Science Course**
R.E. Ladner, **A. Oleson**, A.J. Ko (2026)
Special Session @ SIGCSE 2026
<https://doi.org/10.1145/3770761.3777138>
- [w1] **EduCHI 2023: 5th Annual Symposium on HCI Education**
C. Gray, C.M. MacDonald, C. Lallemand, **A. Oleson**, A.R.L. Carter, O. St-Cyr, C. Pitt (2023)
EduCHI Annual Symposium on HCI Education @ CHI 2023
<https://doi.org/10.1145/3544549.3573790>

Workshop Papers

- [a4] **Software Developer Diversity and Inclusion Workshop**
Organizers: D. Ford, E. Murphy-Hill, M. Storey, R. Prikladnicki, Y. Wang
National Institute of Informatics (NII) Shonan Meeting (2023)
In the style of Dagstuhl seminars, this invited workshop brought together leading researchers doing diversity and inclusion work in software engineering to identify challenges and gather best practices.
<https://shonan.nii.ac.jp/seminars/194/>
- [a3] **Toward the Development of HCI Pedagogical Content Knowledge**
A. Oleson, A.J. Ko (2020)
EduCHI Annual Symposium on HCI Education @ CHI'20
<https://educhi2020.educhi.org/paper/toward-the-development-of-hci-pedagogical-content-knowledge/>

- [a2] **The GenderMag-Teach Project**
M.M. Burnett, Z. Steine-Hanson, **A. Oleson** (2019)
EduCHI Annual Symposium on HCI Education @ CHI'19
<https://web.engr.oregonstate.edu/~burnett/Reprints/eduCHI19-GenderMagTeach-preprint.pdf>
- [a1] **Gender Biases in Software for Problem-Solving**
M.M. Burnett, A. Sarma, C. Mendez, **A. Oleson**, C. Hilderbrand, Z. Steine-Hanson, A.J. Ko (2018)
Designing Technologies to Support Human Problem Solving @ VL/HCC'18
<https://alannaholeson.com/papers/vlhcc18-workshopPaper-problemSolving.pdf>

Patents

- [p1] **Smart Guide to Capture Digital Images that Align with a Target Image Model**
A. Oleson, R. Mech, J. Echevarria, C. Lu (2018)
Filed as US15/897,951; published as US20190253614A1 and US10574881B2
<https://patents.google.com/patent/US10574881B2/en>

Dissertation

- [t1] **Integrating Inclusive Design and Computing Education**
A. Oleson (2023)
For a Ph.D. in Information Science at the University of Washington
<https://www.proquest.com/docview/2915914933/F660E83EEF534C24PQ/3>

Doctoral Consortia Submissions (Archived)

- [d2] **CIDER: A Method to Teach Practical Critical Software Design Skills**
A. Oleson (2022)
SIGCSE DC @ ACM ICER Conference on International Computing Education Research
<https://doi.org/10.1145/3501709.3544295>
- [d1] **Supporting Critical Software Design Decision-Making in HCI Education**
A. Oleson (2022)
DUB DC @ University of Washington

Digital Books

These books are free, online, accessible, and responsive resources for the computing education community. They are not peer-reviewed in the traditional sense, but instead iteratively improved by feedback from community members in response to the rapidly evolving computing field. I am continuously improving and expanding them.

- [d4] **Teaching Accessible Computing**, 2nd ed. (2026)
A. Oleson, A.J. Ko, R. Ladner
10,000+ readers in 115 countries
<https://bookish.press/tac/>
Adds three new chapters on integrating accessibility content into Operating Systems, Sustainable Computing, and Capstone/Thesis courses in postsecondary CS curricula.

- [d3] **Critically Conscious Computing: Methods for Secondary Education**, 2nd ed. (2026)
A.J. Ko, A. Beitlers, B. Wortzman, M. Davidson, **A. Oleson**, M. Kirdani-Ryan, S. Druga, J. Everson
<https://criticallyconsciouscomputing.org/>
Updated to reflect the CS Teachers Association (CSTA) revised [PK-12 CS learning standards](#). The 1st edition of the book strongly impacted the revision process and new standards with its recommendations, which will likely be adopted in all 50 US states.
- [d2] **Teaching Accessible Computing**, 1st ed. (2024)
A. Oleson, A.J. Ko, R. Ladner
<https://bookish.press/tac/>
Offers foundational knowledge about disability and pedagogy to support CS faculty who want to teach about accessibility in their courses, but don't know where to start. The 1st edition of this book contains perspectives from 21 domain experts on how to integrate accessibility into 14 CS courses (intro programming, graphics, data structures, etc.).
- [d1] **Critically Conscious Computing: Methods for Secondary Education**, 1st ed. (2022)
A.J. Ko, A. Beitlers, B. Wortzman, M. Davidson, **A. Oleson**, M. Kirdani-Ryan, S. Druga, J. Everson
<https://criticallyconsciouscomputing.org/>
A book for secondary educators who want to teach CS from a more critical lens, examining the discipline and its content knowledge from technical, sociotechnical, and sociopolitical stances.
CS and Design (chapter in Critically Conscious Computing)
Authors: A.J. Ko, **A. Oleson**
<https://criticallyconsciouscomputing.org/design>
Describes how design and computing intersect in educational contexts, including the differences between design and engineering mindsets, different pedagogical approaches, and a unit sketch for teaching CS as design-informed engineering.

Research Professional Development

- Collaborative Institutional Training Initiative (**CITI Program**) Certifications:
Human Subjects Research; Responsible Conduct of Research for Social, Behavioral, & Educational Research; Conflicts of Interest; Advanced Research Security; Undue Foreign Influence Risks & Mitigations
- National Science Foundation (NSF) **CISE CAREER Workshop** (May 2026)
- MasterClass: **Getting Published with Nature Journals** Webinar (May 2026)
- **Federal Grant Writing Workshop**, U. Denver (Summer 2025)
- **New Faculty in STEM Mentoring Program**, U. Denver (2024-2025)

Teaching Experience

- Assistant Professor **Design Justice**, COMP 3701 @ U. Denver
New course, original design. Seminar. Students read about, discuss, and reflect upon what “justice” means in today’s technological landscape from different theoretical perspectives (e.g. disability justice, anti-racist design), articulating and refining their values and commitments as tomorrow’s technology practitioners.
- User Experience Design Methods**, COMP 3705 @ U. Denver
New course, original design. Flipped classroom. Introduces students to the basic methods needed to succeed in UX design careers, such as problem framing, ideation methods, low- and mid-fidelity prototyping, design evaluation methods, and synthesis of user research. Students deeply explore a design problem of interest from multiple perspectives and present a design specification at the end of the term.

Human-Computer Interaction, COMP 3/4100 @ U. Denver

Substantially updated design. Introductory HCI/UX design course redesigned to integrate active learning and discussion activities in every class session, based on effective HCI education research. Students learn basic HCI principles and design methods through digital prototyping, then build collaboration skills through a final design team project to address a structural inequity in their communities.

Introduction to Computer Science, COMP 1201 @ U. Denver

Designed and implemented (in collaboration with colleagues) an introductory CS course (required for majors/minors, open to all) on algorithmic thinking, career readiness, and ethical perspectives on technology. Students engage in activities and discussions to build CS foundations and identities.

Instructor of **User-Centered Design Methods, INFO 360 @ U. Washington**

Record *Adapted course materials to support more equitable participation during COVID-19 remote learning; Created original assignments to help students recognize and respond to exclusionary design biases.*

Teaching **Cooperative Software Development, INFO 442 @ U. Washington**

Assistant *Helped adapt course content to equitable remote instruction formats, focusing on needs of the most disadvantaged student groups first, and implemented structures for instructor and peer support.*

User-Centered Design Methods, INFO 360 @ U. Washington

Created formative assessment questions to determine gaps in students' understandings and adapt instruction accordingly; Prepared and delivered lesson on interface evaluation methods.

Guest **Computing Education Research Seminar, CSE 599 @ U. Washington**

Instructor *Led discussion on the idea of "CS For All" and how to equitably expand CS education access.*

Design Methods, INFO 360 @ U. Washington

Presented an overview of the principles of various empirical and analytical user interface evaluation methods and led a series of active learning activities for students to engage with each method.

Teaching Professional Development

- **Advancing Curriculum with Entrepreneurial Mindset (ACE) Workshop, KEEN Program (July 2026)**
- **Teaching & Learning for Regeneration & Resilience Workshop, Center for Regenerative Futures, U. Denver (November 2025)**
- **Mastery-Based Learning Workshop, U. Denver (August 2025)**
- **Designing Courses For and With AI Workshop, U. Denver (December 2024)**
- **Universal Design for Learning (UDL) Training for Faculty, U. Denver (Fall 2024)**
- **Being a Holistic Educator: Teacher-Scholar Lecture Series, U. Denver (Fall 2024)**
- **New Faculty Academy, U. Denver (2024-2025)**
- **Teaching Practicum I & II, U. Washington Information School (2019-2020)**

Students Supervised & Mentored

I have meaningfully mentored 29 students over the course of my career (11 grad, 16 undergrad, 2 high school), including several women, non-binary, and transgender people, many Students of Color, several international students, a few from rural areas, and several students from economically disadvantaged backgrounds, with varied (dis)abilities and prior exposure to academia. 16 of these students have co-authored academic papers with me. At least 13 of the undergraduates and high schoolers chose to pursue graduate education after working with me. Below, I list only formal advisees to preserve privacy of my less formal mentees who seek my advice and support.

Research Advisees

Ph.D. Students

Faiza Tazi, Computer Science @ U. Denver 2024 - 2026
Dissertation: *Beyond One-Size-Fits-All: Privacy, Security, and Stigma in Telehealth Through the Lens of Marginalized Users*

Suleiman Saka, Computer Science @ U. Denver 2024 - present
Dissertation: *Empowering Older Adults' Privacy Decisions in Healthcare IoT: An LLM-Based System Grounded in Human-Centered Design Principles*

Sunny Shrestha, Computer Science @ U. Denver 2024 - present
RA Project: *Multisite Comparison of Embedding Socially Responsible Design Across the Undergraduate Computer Science Curriculum*

Master's Students

Yuliya Kim, Computer Science @ U. Denver 2025 - present
Thesis: *Foundations for Regenerative AI Education in Humanities and Social Science Courses (working title)*

Undergraduate Students

Pluto Hassan, Computer Science @ U. Denver 2025 - present
Thesis: *TBD*

Muskan Fatima, Computer Science & Mathematics @ U. Denver 2024 - 2026
Thesis: *First Impressions Matter: How CS Syllabi Communicate and Reflect Accessibility Norms*

Mars Kaplan, Computer Science & Spanish @ U. Denver 2024 - 2025

Olivia Nagy, Psychology @ U. Denver 2024 - 2025

Anael Kuperwajs Cohen, REU @ U. Washington 2023 - 2024

Audrey Le Meur, REU @ U. Washington 2022 - 2022

Lena Armstrong, REU @ U. Washington 2022 - 2022

F. Megumi Kivuva, REU @ U. Washington 2022 - 2022

Christopher Perdriau, REU @ U. Washington 2020 - 2021

Meron Solomon, Research Assistant @ U. Washington 2019 - 2021

Christopher Perdriau, Research Assistant @ Oregon State 2016 - 2018

Zoë Steine-Hanson, Research Assistant @ Oregon State 2016 - 2018

Committee Memberships

Dissertation Committees

Sunny Shrestha, Computer Science @ U. Denver TBD
Dissertation: *Gender Bias, AI Tools, and Public Policy Community Education*
Advisor: Benjamin Xie

Ericka Corral , Computer Science @ U. Denver Dissertation: <i>Community-centered Environmental AI Justice</i> Advisor: Benjamin Xie	TBD
Demi Jaiyeola , Computer Science @ U. Denver Dissertation: <i>Exploring Neurodiverse Programmers' Usage of AI Tools</i> Advisor: Benjamin Xie	TBD
Yajing Li , Computer Science @ U. Denver Dissertation: TBD Advisor: Benjamin Xie	TBD
Jordan Sinclair , Computer Science @ U. Denver Dissertation: <i>Multimodal Preference Learning from Human Feedback</i> Advisor: Mohammad Mahoor	TBD 2027
Ren Butler , Human-Computer Interaction @ Carnegie Mellon University Dissertation: <i>Supporting Neurodiverse Communication in AI-Driven Software Engineering</i> Advisor: Andrew Begel	TBD 2026
Parthasarathay PD , Computer Science @ Birla Institute of Technology & Science, Pilani Dissertation: <i>ALPHA: A Design-Based Research Approach to Game-Based Learning for Digital Accessibility Education</i> Advisor: Swaroop Joshi	TBD 2026
Jixiang Fan , Computer Science @ Virginia Tech Dissertation: <i>From Diary to Discussion: The PCAR Framework for Scaffolding Empathy-Oriented Learning in HCI Education</i> Advisor: Scott McCrickard	July 2026

Master's Committees

Maisey Toczek , Computer Science @ U. Denver Thesis: <i>Brush-E Bot: Exploring the Benefits of a Robot Companion to Improve Children's Oral Hygiene Habits</i> Advisor: Kerstin Haring	August 2025
---	-------------

Other Degree Milestone Committees

Jordan Sinclair , Computer Science @ U. Denver Qualifying Exam <i>Preference Learning from Human Feedback</i> Advisor: Mohammad Mahoor	August 2026
Amjad Alghamdi , Computer Science @ U. Denver Qualifying Exam <i>Augmented and Virtual Reality for Children with Special Needs: Systems, Sensing, Adaptation, and Ethics, with a Focus on Autism Spectrum Disorder</i> Advisor: Mohammad Mahoor	June 2026

Weston Laity, Computer Science @ U. Denver

May 2025

Qualifying Exam

Ethical, Statistical, Empirical, and Theoretical Foundations to Inform Mind

Migration in Human-Robot Interaction

Advisor: Kerstin Haring

Sunny Shrestha, Computer Science @ U. Denver

May 2025

Qualifying Exam

Algorithmic Fairness and Gender Bias in Recommender Systems

Advisor: Chris Gauthier-Dickey

Professional Memberships

Association for Computing Machinery

AccessComputing

ACM Special Interest Group on Computer-Human Interaction (SIGCHI)

ACM Special Interest Group on Computer Science Education (SIGCSE)

Professional Reviewing

Journals **ACM TOCE**: Transactions on Computing Education

ACM TOSEM: Transactions on Software Engineering Methodology

International Journal of Child-Computer Interaction (**CCI**)

Conferences **ACM CHI** Conference on Human Factors in Computing Systems

ACM CSCW Conference on Computer-Supported Cooperative Work and Social Computing

ACM EduCHI Symposium on HCI Education

ACM SIGCSE Technical Symposium on Computer Science Education

IEEE VL/HCC Conference on Visual Languages & Human-Centered Computing

Publishers **MIT Press**

Professional Service

Steering **EduCHI Symposium on HCI Education** (2023-present)

Committee *Providing oversight and guidance to the organizing committee during the transition to a standalone conference in 2024; Liaison to ACM TOCE journal and co-director for early-career researcher workshops.*

Partner **AccessComputing**, University of Denver (2024-present)

Acting as DU's liaison to the AccessComputing program to support participation and access for computing students with disabilities; Connecting DU faculty and staff to accessible computing resources.

Associate **ACM Transactions on Computing Education** (2024-2025)

Editor *Expertise areas: Human-computer interaction (HCI) education and computing design education research.*

Technical **EduCHI Symposium on HCI Education** (2023, 2025)

Program Chair *Oversaw the paper review process for the symposium's research and teaching-related submissions and worked with program committee to make final acceptance decisions; Collaborated with the General Chairs to create and promote community-building workshops prior to the symposium.*

- Departmental Service @ U. Denver
- **Tenure-Track Hiring Committee**, 2025-2026
 - **Graduate Admissions Restructuring Committee**, 2025-2027
 - **Graduate Curriculum Committee**, 2024-2025
 - **Undergraduate Honors, Awards, and Distinction Committee**, 2024-2025
- School Service @ U. Denver
- **Academic Appeal Committee** (ad hoc), 2025
- Discussant
- RESPECT Conference Doctoral Consortium** (2025)
- Discussant
- Humanization of Computing and Engineering Education Doctoral Consortium** (2025)
- Organizer
- HCI Pedagogy Workshop**, EduCHI Symposium (2024)
Planned and led the inaugural half-day workshop on HCI pedagogy and pedagogical research for doctoral students, postdocs, and 1st year faculty, which became a recurring event due to its success.
- Mentor
- Trans/Queer in HCI Mentorship Program** (2022-2023)
Providing professional and personal support to students in the Human-Computer Interaction field who are part of the LGBTQIA2S+ community.
- Chair
- UW iSchool Doctoral Student Association** (2021-2022)
Organized and presided at DSA meetings and town halls; Led DSA outreach at new student orientation; Advocated for student perspectives to faculty, staff, and administrators.
- Officer
- Secretary**, UW iSchool Doctoral Student Association (2020-2021)
Organized and disseminated information internally to doctoral students, including meeting minutes; oversaw internal DSA officer elections.
- Communications & Outreach**, UW iSchool Doctoral Student Association (2019-2020)
Administration of internal \& external communications: mailing lists, social media accounts, reports on DSA activity, other outreach channels as requested.
- Student Volunteer
- ACM **CHI** Conference (2019, 20, 22, 23), ACM **SIGCSE** Technical Symposium (2019)
 ★ *CHI 2019: Received internal SV award "for going above and beyond" typical SV duties to ensure conference participants' accessibility needs were properly met during paper sessions.*
- Student Coordinator
- UW DUB Seminar** (2019-2020)
DUB (Design, Use, Build) is an interdisciplinary community at the UW focused on HCI and Design which coordinates weekly invited talks with internal and external speakers.
- Peer Mentor
- OSU STEM Leaders Program** (2015-2018)
Helped 5 freshmen from underrepresented backgrounds in STEM transition to and succeed in new college environments as they completed original research projects with faculty mentors.